

**EGERTON**



**UNIVERSITY**

**UNIVERSITY EXAMINATIONS**

**REGULAR -NJORO CAMPUS**

**SECOND SEMESTER, 2021/2022 ACADEMIC YEAR**

**THIRD YEAR RE-SIT/SPECIAL EXAMINATION FOR THE DEGREE OF  
BACHELOR OF SCIENCE IN ENVIRONMENTAL SCIENCE**

**ENSC 306/333: ENVIRONMENTAL HEALTH**

**STREAM: B.Sc. (ENSC & NARE)**

**TIME: 2 HRS**

**EXAMINATION SESSION: MAY**

**YEAR: 2022**

**INSTRUCTIONS:**

- (i) Attempt all questions in Section A and any TWO in Section B
- (ii) Marks for each question are indicated in parenthesis
- (iii) Do not write on the question paper
- (iv) Illustrate your answers with suitable diagrams and formulas where appropriate

**SECTION A (40 MARKS)**

**Question One**

Write short notes on the following terms as used in environmental health.

- a) The inner versus the outer environment **(4 Marks)**
- b) Chemical interactions **(3 Marks)**
- c) Selective toxicity **(3 Marks)**

**Question Two**

Write short notes on the following terms as used in environmental health.

- a) Many factors impact sensitivity to a pollutant in human beings. Explain these factors and give examples in each case. **(4 Marks)**

- b) What do you understand by environmental hormones? Discuss the impacts of environmental hormones to fish and other wildlife (6 Marks)

**Question Three**

- a) Outline and explain the primary methods that have proved to be effective for preserving food (4 Marks)
- b) Name any two gaseous indoor pollutants, their sources and the acute effects/ symptoms associated with each. (6 Marks)

**Question Four**

All cells possess an inherent rate of cellular division, one in which the rate of normal death cell is matched by the formation of new cells. Discuss in detail how this proliferation of cells develops to form cancer. (10 Marks)

**SECTION B (30 MARKS)**

**Question Five**

"The specific response that an organism exhibits from a given dose of a toxin varies". Discuss (15 Marks)

**Question Six**

There are common toxins introduced into food through improper handling. Discuss the main organisms that contaminate food as a result of improper handling. (15 Marks)

**Question Seven**

Dose-response assessment is an important step in risk assessment with respect to environmental toxicants. Discuss the various steps to be considered in dose-response assessment. (15 Marks)

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